



UNIVERSITÀ DEGLI STUDI DI PISA

CORSO DI STUDIO IN BIONICS ENGINEERING

STATISTICAL SIGNAL PROCESSING - *Prova scritta preliminare del 6/02/2020*

PROBLEM

We observe N samples (for $n=0, 1, \dots, N-1$, with $N \gg 1$) of a realization of the random process $X[n] = s[n] + W[n]$, where $s[n] = A \cos(2\pi F_0 n) + B \cos^2(2\pi F_0 n)$ is the signal of interest (SOI), A and B are the amplitudes of the SOI modelled as unknown deterministic parameters, and $W[n]$ is a zero-mean white Gaussian process with power σ_w^2 .

- (1) Derive the signal to noise ratio (SNR) $\gamma \triangleq E\{\|\mathbf{s}\|^2\} / E\{\|\mathbf{W}\|^2\}$, where $\mathbf{s}$ and $\mathbf{W}$ are the vectors containing the N samples of signal and noise, respectively. Then, derive the log-likelihood function $LLF(A, B, \sigma_w^2) = \ln f_X(\mathbf{x}; A, B, \sigma_w^2)$, where $f_X(\mathbf{x}; A, B, \sigma_w^2)$ is the probability density function (PDF) of the observed data vector $\mathbf{X} = [X[0] \ X[1] \ \dots \ X[N-1]]^T$.
- (2) Derive the joint maximum likelihood (ML) estimators of the signal amplitudes A and B when the noise power σ_w^2 is known and provide a geometrical interpretation of them.
- (3) Derive the bias and the mean square error (MSE) of the ML estimators of A and B , establish if they are unbiased and consistent or not, plot the MSE of $\hat{B}_{ML}$ as a function of N and find the minimum value of N which guarantees an MSE lower than 10^{-3} when the noise power is $\sigma_w^2 = 1$.
- (4) Find the Cramér-Rao bounds (CRBs) on the joint estimation of A and B and establish if the ML estimators previously derived are efficient or not.
- (5) Assume now that also the noise power σ_w^2 is unknown: derive an unbiased estimator of σ_w^2 .



UNIVERSITÀ DEGLI STUDI DI PISA

CORSO DI STUDIO IN BIONICS ENGINEERING

Solution.

(1) Derive the SNR and the log-likelihood function.

$$s[n] = A \cos(2\pi F_0 n) + B \cos^2(2\pi F_0 n) \Rightarrow \mathbf{s} = A\mathbf{h}_1 + B\mathbf{h}_2 = [\mathbf{h}_1 \quad \mathbf{h}_2] \begin{bmatrix} A \\ B \end{bmatrix} = \mathbf{H}\boldsymbol{\theta}$$

$$E\{\|\mathbf{s}\|^2\} = \|\mathbf{s}\|^2 = \|A\mathbf{h}_1 + B\mathbf{h}_2\|^2 = A^2 \|\mathbf{h}_1\|^2 + B^2 \|\mathbf{h}_2\|^2, \quad \text{since } \mathbf{h}_1^T \mathbf{h}_2 = 0, \quad \text{i.e. } \mathbf{h}_1 \perp \mathbf{h}_2,$$

In fact,

$$\begin{aligned} \mathbf{h}_1^T \mathbf{h}_2 &= \sum_{n=0}^{N-1} \cos(2\pi F_0 n) \cdot \cos^2(2\pi F_0 n) = \sum_{n=0}^{N-1} \cos(2\pi F_0 n) \left[\frac{1}{2} + \frac{1}{2} \cos(4\pi F_0 n) \right] \\ &= \frac{1}{2} \sum_{n=0}^{N-1} \cos(2\pi F_0 n) + \frac{1}{2} \sum_{n=0}^{N-1} \cos(2\pi F_0 n) \cdot \cos(4\pi F_0 n) \\ &= \frac{1}{2} \sum_{n=0}^{N-1} \cos(2\pi F_0 n) + \frac{1}{4} \sum_{n=0}^{N-1} \cos(6\pi F_0 n) + \frac{1}{4} \sum_{n=0}^{N-1} \cos(2\pi F_0 n) \cong 0 \quad (\text{if } N \gg 1) \end{aligned}$$

$$E\{\|\mathbf{s}\|^2\} = A^2 \|\mathbf{h}_1\|^2 + B^2 \|\mathbf{h}_2\|^2 = \frac{A^2 N}{2} + \frac{B^2}{4} \left(N + \frac{N}{2} \right) = \frac{N}{2} \left(A^2 + \frac{3B^2}{4} \right), \quad E\{\|\mathbf{W}\|^2\} = N\sigma_w^2,$$

$$\gamma \triangleq E\{\|\mathbf{s}\|^2\} / E\{\|\mathbf{W}\|^2\} = \frac{1}{2\sigma_w^2} \left(A^2 + \frac{3B^2}{4} \right)$$

$$X[n] \in N(s[n], \sigma_w^2), \quad \{X[n]\}_{n=0}^{N-1} \text{ independent} \Rightarrow \mathbf{X} \in N(\mathbf{H}\boldsymbol{\theta}, \sigma_w^2 \mathbf{I})$$

PDF in scalar format:

$$\begin{aligned} f_X(\mathbf{x}; A, B, \sigma_w^2) &= \prod_{n=0}^{N-1} f_X(x[n]; A, B, \sigma_w^2) \\ &= \left(\frac{1}{\sqrt{2\pi\sigma_w^2}} \right)^N \exp \left(-\frac{1}{2\sigma_w^2} \sum_{n=0}^{N-1} (x[n] - A \cos(2\pi F_0 n) - B \cos^2(2\pi F_0 n))^2 \right) \end{aligned}$$

PDF in vector format:

$$\begin{aligned} f_X(\mathbf{x}; A, B, \sigma_w^2) &= \frac{1}{(2\pi)^{N/2} \sqrt{|\sigma_w^2 \mathbf{I}|}} \exp \left(-\frac{1}{2} (\mathbf{x} - \mathbf{H}\boldsymbol{\theta})^T (\sigma_w^2 \mathbf{I})^{-1} (\mathbf{x} - \mathbf{H}\boldsymbol{\theta}) \right) \\ &= \frac{1}{(2\pi)^{N/2} (\sigma_w^2)^{N/2}} \exp \left(-\frac{1}{2\sigma_w^2} \|\mathbf{x} - \mathbf{H}\boldsymbol{\theta}\|^2 \right) \end{aligned}$$



UNIVERSITÀ DEGLI STUDI DI PISA

CORSO DI STUDIO IN BIONICS ENGINEERING

The log-likelihood function is given by:

$$LLF(A, B, \sigma_w^2) = \ln f_X(\mathbf{x}; A, B, \sigma_w^2) = -\frac{N}{2} \ln(2\pi) - \frac{N}{2} \ln(\sigma_w^2) - \frac{1}{2\sigma_w^2} \|\mathbf{x} - \mathbf{H}\boldsymbol{\theta}\|^2$$

(2) Derive the joint maximum likelihood (ML) estimators of the signal amplitudes A and B when the noise power σ_w^2 is a priori known and provide a geometrical interpretation of them.

$$\hat{\boldsymbol{\theta}}_{ML} = (\mathbf{H}^T \mathbf{H})^{-1} \mathbf{H}^T \mathbf{X} = \begin{bmatrix} \|\mathbf{h}_1\|^2 & \mathbf{h}_1^T \mathbf{h}_2 \\ \mathbf{h}_1^T \mathbf{h}_2 & \|\mathbf{h}_2\|^2 \end{bmatrix}^{-1} \begin{bmatrix} \mathbf{h}_1^T \\ \mathbf{h}_2^T \end{bmatrix} \mathbf{X} = \begin{bmatrix} \frac{N}{2} & 0 \\ 0 & \frac{3N}{8} \end{bmatrix}^{-1} \begin{bmatrix} \mathbf{h}_1^T \\ \mathbf{h}_2^T \end{bmatrix} \mathbf{X} = \begin{bmatrix} \frac{2}{N} \mathbf{h}_1^T \mathbf{X} \\ \frac{8}{3N} \mathbf{h}_2^T \mathbf{X} \end{bmatrix}$$

$$\hat{A}_{ML} = \frac{2}{N} \mathbf{h}_1^T \mathbf{X} = \frac{2}{N} \sum_{n=0}^{N-1} x[n] \cos(2\pi F_0 n), \quad \hat{B}_{ML} = \frac{8}{3N} \mathbf{h}_2^T \mathbf{X} = \frac{8}{3N} \sum_{n=0}^{N-1} x[n] \cos^2(2\pi F_0 n)$$

A and B are estimated at the output of two (orthogonal) linear matched filters: they are the components of the received signal $\mathbf{X}$ onto the signal subspace, i.e. the subspace generated by the columns of $\mathbf{H}$. The two estimators are linear, so they are Gaussian distributed.

(3) Derive the bias and the mean square error (MSE) of the ML estimators of A and B , establish if they are unbiased and consistent or not, plot the MSE of $\hat{B}_{ML}$ as a function of N and find the minimum value of N which guarantees an MSE lower than 10^{-3} when the noise power is $\sigma_w^2 = 1$.

$$E\{\hat{A}_{ML}\} = \frac{2}{N} \mathbf{h}_1^T E\{\mathbf{X}\} = \frac{2}{N} \mathbf{h}_1^T (A\mathbf{h}_1 + B\mathbf{h}_2) = A$$

$$E\{\hat{B}_{ML}\} = \frac{8}{3N} \mathbf{h}_2^T E\{\mathbf{X}\} = \frac{8}{3N} \mathbf{h}_2^T (A\mathbf{h}_1 + B\mathbf{h}_2) = B$$

The estimators are unbiased.

$$\begin{aligned} MSE\{\hat{A}_{ML}\} &= E\left\{\left(A - \hat{A}_{ML}\right)^2\right\} = E\left\{\left(A - \frac{2}{N} \mathbf{h}_1^T \mathbf{X}\right)^2\right\} = E\left\{\left(A - \frac{2}{N} \mathbf{h}_1^T (A\mathbf{h}_1 + B\mathbf{h}_2 + \mathbf{W})\right)^2\right\} \\ &= E\left\{\left(A - A - \frac{2}{N} \mathbf{h}_1^T \mathbf{W}\right)^2\right\} = E\left\{\left(-\frac{2}{N} \mathbf{h}_1^T \mathbf{W}\right)^2\right\} = \frac{4}{N^2} E\left\{\mathbf{h}_1^T \mathbf{W} (\mathbf{h}_1^T \mathbf{W})^T\right\} \\ &= \frac{4}{N^2} E\left\{\mathbf{h}_1^T \mathbf{W} \mathbf{W}^T \mathbf{h}_1\right\} = \frac{4\sigma_w^2}{N^2} \mathbf{h}_1^T \mathbf{h}_1 = \frac{2\sigma_w^2}{N} \end{aligned}$$

$$MSE\{\hat{B}_{ML}\} = \frac{8\sigma_w^2}{3N}$$



UNIVERSITÀ DEGLI STUDI DI PISA

CORSO DI STUDIO IN BIONICS ENGINEERING

The estimators are consistent.

$$\sigma_w^2 = 1: \quad MSE\{\hat{B}_{ML}\} = \frac{8}{3N} \leq 10^{-3} \quad \Rightarrow \quad N \geq \frac{8000}{3} = 2667$$

Note that the derivative of the logarithm of the LR can be put in the following form:

$$\ln f_x(\mathbf{x}; A, B, \sigma_w^2) = -\frac{N}{2} \ln(2\pi) - \frac{N}{2} \ln(\sigma_w^2) - \frac{1}{2\sigma_w^2} ((\mathbf{x} - \mathbf{H}\boldsymbol{\theta})^T (\mathbf{x} - \mathbf{H}\boldsymbol{\theta}))$$

$$\frac{\partial \ln L(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}} = \frac{1}{\sigma_w^2} [\mathbf{H}^T (\mathbf{x} - \mathbf{H}\boldsymbol{\theta})] = \frac{1}{\sigma_w^2} [\mathbf{H}^T \mathbf{x} - \mathbf{H}^T \mathbf{H} \boldsymbol{\theta}] = \frac{(\mathbf{H}^T \mathbf{H})}{\sigma_w^2} [(\mathbf{H}^T \mathbf{H})^{-1} \mathbf{H}^T \mathbf{x} - \boldsymbol{\theta}] = \mathbf{I}(\boldsymbol{\theta}) [\mathbf{g}(\mathbf{x}) - \boldsymbol{\theta}]$$

$$\left\{ \begin{array}{l} FIM : \quad \mathbf{I}(\boldsymbol{\theta}) = \frac{(\mathbf{H}^T \mathbf{H})}{\sigma_w^2} \rightarrow CRB(\boldsymbol{\theta}) = \sigma_w^2 (\mathbf{H}^T \mathbf{H})^{-1} = \sigma_w^2 \begin{bmatrix} \frac{N}{2} & 0 \\ 0 & \frac{3N}{8} \end{bmatrix}^{-1} = \begin{bmatrix} \frac{2\sigma_w^2}{N} & 0 \\ 0 & \frac{8\sigma_w^2}{3N} \end{bmatrix} \\ \hat{\boldsymbol{\theta}}_{ML} = \mathbf{g}(\mathbf{x}) = (\mathbf{H}^T \mathbf{H})^{-1} \mathbf{H}^T \mathbf{x} = \mathbf{H}^\# \mathbf{x} \end{array} \right.$$

Hence, the estimator is unbiased and efficient, in addition to be linear, Gaussian, and consistent.

$$MSE(\hat{A}_{ML}) = CRB(A) = \frac{2\sigma_w^2}{N}, \quad MSE(\hat{B}_{ML}) = CRB(B) = \frac{8\sigma_w^2}{3N}$$

(4) Find the Cramér-Rao bounds (CRBs) on the joint estimation of A and B and establish if the ML estimators previously derived are efficient or not.

The FIM is as above. It could have been derived by calculating the 2nd order derivatives:

$$-E \left\{ \frac{d^2 \ln f_x(\mathbf{x})}{dA^2} \right\} = \frac{N}{2\sigma_w^2}, \quad -E \left\{ \frac{d^2 \ln f_x(\mathbf{x})}{dB^2} \right\} = \frac{3N}{8\sigma_w^2}, \quad -E \left\{ \frac{d^2 \ln f_x(\mathbf{x})}{dA dB} \right\} = 0$$

(5) Assume now that also the noise power σ_w^2 is unknown: derive an unbiased estimator of σ_w^2 .



UNIVERSITÀ DEGLI STUDI DI PISA

CORSO DI STUDIO IN BIONICS ENGINEERING

$$\frac{dLLF(\hat{A}_{ML}, \hat{B}_{ML}, \sigma_w^2)}{d\sigma_w^2} = \frac{d \ln f_X(\mathbf{x}; \hat{A}_{ML}, \hat{B}_{ML}, \sigma_w^2)}{d\sigma_w^2} = -\frac{N}{2\sigma_w^2} + \frac{1}{2\sigma_w^4} \|\mathbf{x} - \mathbf{H}\hat{\boldsymbol{\theta}}_{ML}\|^2 = 0$$

$$\hat{\sigma}_{w,ML}^2 = \frac{1}{N} \|\mathbf{x} - \mathbf{H}\hat{\boldsymbol{\theta}}_{ML}\|^2 = \frac{1}{N} \|\mathbf{x} - \mathbf{H}\mathbf{H}^H \mathbf{x}\|^2 = \frac{1}{N} \|(\mathbf{I} - \mathbf{H}\mathbf{H}^H) \mathbf{x}\|^2 = \frac{1}{N} \|(\mathbf{I} - \mathbf{P}_H) \mathbf{x}\|^2 = \frac{1}{N} \mathbf{x}^T (\mathbf{I} - \mathbf{P}_H) \mathbf{x}$$

The ML estimator projects the received data vector $\mathbf{X}$ onto the subspace orthogonal to the signal subspace and then calculates the energy of the data after this projection, i.e. the sum of the powers of the $N-2$ components where only the noise is present. Since the signal subspace is 2-D, the orthogonal noise subspace is $(N-2)$ -D, hence the mean $E\{\mathbf{x}^T (\mathbf{I} - \mathbf{P}_H) \mathbf{x}\}$ is equal to $(N-2)\sigma_w^2$.

Hence, if we want to have an unbiased estimator we should divide by $N-2$ instead of N :

$$\hat{\sigma}_{w,unbiased}^2 = \frac{1}{N-2} \mathbf{x}^T (\mathbf{I} - \mathbf{P}_H) \mathbf{x}$$